Sport Sciences for Health (2025) 21:3189–3196
https://doi.org/10.1007/s11332-025-01529-0

RESEARCH

CrossFit vs. sedentary lifestyle: unveiling the effects on strength, balance and quality of life

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Received: 1 July 2025 / Accepted: 12 August 2025 / Published online: 26 September 2025

Abstract
Background CrossFit is a high-intensity training modality that combines functional exercises targeting strength, endurance and flexibility. In contrast to a sedentary lifestyle, CrossFit has been associated with both physical and psychological benefits; however, its impact on biomechanical parameters and quality of life remains to be fully explored.
Objective To compare gait kinetics, postural balance and quality of life between young adults who practise CrossFit and those with a sedentary lifestyle.
Methods A cross-sectional observational study was conducted involving 60 participants (30 CrossFit practitioners and 30 sedentary individuals), aged between 25 and 35 years (mean age 27.55 ± 3.03). Ground reaction forces during gait and balance (Romberg test with eyes open and closed) were assessed using force platforms (Dinascan/IBV® P600). Quality of life was evaluated using the validated SF-36 and WHOQOL-BREF questionnaires.
Results Although no significant differences were observed in ground reaction forces ($p \geq 0.140$), the CrossFit group exhibited greater walking speed and stance time. During the eyes-open Romberg test, the CrossFit group demonstrated greater anteroposterior sway ($p = 0.019$). Quality of life scores were significantly higher in the CrossFit group ($p \leq 0.009$).
Conclusion CrossFit is associated with improvements in perceived quality of life and certain aspects of postural balance, highlighting its potential as an effective tool to promote active and healthy lifestyles in contrast to sedentary behaviour. Further research with greater statistical power is warranted.

Keywords High-intensity interval training · Sedentary behaviour · Gait analysis · Postural balance · Quality of life

Introduction

CrossFit is a high-intensity exercise modality that integrates elements of weightlifting, gymnastics and metabolic conditioning, designed to develop broad physical capacity through varied and functional routines [1–3]. It originated in the 1990 s under the direction of Greg Glassman in California, with a philosophy grounded in safety, efficiency and efficacy, aiming to optimise physical competence across ten recognised fitness domains. This practice has gained global popularity not only for its benefits in increasing strength, endurance and cardiovascular capacity, but also for its positive impact on psychosocial well-being and personal satisfaction [3].

Training sessions, known as the "Workout of the Day" (WODs) [4], are intense and varied, encompassing everything from Olympic lifts to gymnastic movements and cardiovascular endurance exercises [3, 5]. Despite its growing popularity, CrossFit has also sparked debate regarding potential injury risks, owing to the high intensity and technical complexity of some movements. Nevertheless, recent evidence suggests that CrossFit can have a favourable influence on functional capacity, health behaviours and psychosocial well-being, with some studies even reporting gender-specific differences amongst athletes [6]. These benefits, including improvements in body composition and greater adherence to active lifestyles, make CrossFit an appealing option for maintaining and enhancing public health.

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Biomechanical analysis is a key tool for understanding the effects of sports practices such as CrossFit, enabling performance optimisation and injury prevention. Amongst the kinetic methods used in human movement analysis, force platforms occupy a prominent role. Based on Newton's third law (the action–reaction principle) [7], these devices quantify ground reaction forces and are fundamental in gait analysis, stability tests and jump evaluations [8]. Their applications are diverse, ranging from the study of sport techniques to enhance movement efficiency and reduce injury risk, to the assessment of musculoskeletal and neurological disorders, thus facilitating diagnosis, treatment and rehabilitation [9].

Despite the growing body of research on CrossFit and the established use of force platforms in biomechanics, a significant gap remains in the scientific literature concerning a specific and comprehensive biomechanical analysis of CrossFit's impact on postural balance and dynamic stability. Whilst its influence on functional capacity and well-being has been explored [1–3], detailed knowledge about how this training modality affects stabilometric and gait parameters in athletes is still limited. Existing studies tend to focus on performance outcomes or injury prevention [9], without delving into the biomechanical mechanisms underlying balance adaptations. This is particularly relevant given that CrossFit demands a high level of stability and motor control to execute complex and varied movements safely and effectively.

The aim of the present study is to analyse, from a biomechanical perspective and using a force platform, the effects of CrossFit training on postural control and stability in a specific population of practitioners.

The hypothesis is that individuals who practise CrossFit will exhibit better postural control and greater stability in both static and dynamic balance tests compared to a sedentary control group, due to the neuromuscular and proprioceptive demands inherent to this discipline.

Materials and methods

Study design, approvals and inclusion criteria

This study employed an exploratory cross-sectional observational design. A formal sample size calculation was not performed due to the preliminary and exploratory nature of the research; however, data were collected from a representative number of participants to obtain relevant initial findings. The study was approved by the Bioethics and Biosafety Committee of the University of Extremadura (Spain), under registration number 199/2023, which supports its ethical validity.

The study was conducted with CrossFit athletes from Box Cerasus in Plasencia (Cáceres, Spain), as well as with volunteers from the University Centre of Plasencia, University of Extremadura. Participants were classified as sedentary if they engaged in less than 60 min per week of moderate to vigorous physical activity, according to self-reported data based on the National Health Survey [10].

Convenience sampling was used for both the CrossFit group (recruited from Box Cerasus) and the sedentary group (university students). All participants took part voluntarily after providing written informed consent.

It is worth noting that Box Cerasus, a specialised CrossFit training centre from which all participating CrossFit athletes were recruited, demonstrated strong support for the research by allowing us to include their athletes in the study and signing the appropriate informed consent documentation.

Inclusion criteria were: individuals aged between 25 and 35 years; regular CrossFit practitioners with a minimum of 6 months of training experience and at least two sessions per week; sedentary individuals; no existing injuries or pathologies in the lower limbs; and provision of signed informed consent. Individuals practising other types of sports or presenting reduced mobility were excluded from the study.

Sample description

The sample consisted of 60 participants, including 30 CrossFit athletes and 30 sedentary individuals, aged between 25 and 35 years (27.55 ± 3.03). In the CrossFit group, 33.3% (N = 10) were men and 66.7% (N = 20) were women. In the sedentary group, 20% (N = 6) were men and 80% (N = 24) were women. The descriptive characteristics of the sample are presented in Table 1.

Table 1 Sociodemographic and anthropometric characteristics of the sample

Variables	Total (mean ± SD)	CrossFit (mean ± SD)	Sedentary (mean ± SD)
Age (years)	27.55 ± 3.03	29.10 ± 3.28	26.00 ± 1.72
Weight (kg)	70.99 ± 17.06	69.62 ± 16.12	72.36 ± 18.13
Height (m)	1.66 ± 0.08	1.68 ± 0.09	1.64 ± 0.07
BMI (kg/m^2)	21.30 ± 4.83	20.62 ± 4.20	21.97 ± 5.36

SD: standard deviation, *kg* kilogram, *m* metre, *BMI* body mass index

Procedure

Data collection and assessments were conducted at the University Centre of Plasencia, part of the University of Extremadura. Each evaluation was carried out individually and lasted approximately 15 to 20 min.

First, sociodemographic, anthropometric and sports practice data were collected through anamnesis. Participants were then weighed and measured for height. Height was recorded using a wall-mounted measuring tape, with participants standing barefoot, arms extended along the body, and looking straight ahead. Weight was measured using the integrated scale of the force platform (Dinascan/IBV® P600, Biomechanics Institute of Valencia, Spain). The body mass index (BMI) was then calculated [11].

Next, force platform assessments were performed using the Dinascan/IBV® P600 (Fig. 1), developed by the Biomechanics Institute of Valencia, Spain. This platform features two programmes:

- **NedAMH/IBV®,** used to analyse kinetic patterns (the forces applied by the foot on the surface during the stance phase of walking).
- **NedSVE/IBV®,** used to assess balance.

The Dinascan/IBV® P600 platform has prior clinical validation [9]. The platform was calibrated before the start of measurements following the standard procedures recommended by the manufacturer. In addition, each test was repeated three times, and the average of the two most consistent measurements was taken as valid, based on the visual assessment of the kinetic pattern and the absence of errors in foot placement.

The evaluation was divided into two parts: first, ground reaction forces were measured during gait analysis and second, the balance assessment was analysed.

Ground reaction force assessment

Participants were first asked to step onto the platform to record their weight. They were then instructed to walk down the platform's corridor at a normal speed to establish a baseline walking speed before measurements. Following these steps, and according to the protocol outlined by Bus et al. [12], participants were directed to initiate walking with a designated foot, completing a total of eight steps (four left and four right). Each participant received detailed instructions on the correct foot placement on the platform, and measurements were not recorded until they were familiarised with the system and protocol. The measured variables were: average step speed (ASS); average support time (AST); average braking force (ABF); average propulsion force (APF); average take-off force (ATF) and average oscillation force (AOF).

Balance assessment

To assess balance, the subject was first asked to step onto the platform to record their body weight. The upcoming test, the Romberg test, was then explained to the subject [13], along with the required posture for its execution: barefoot, with heels together and toes forming a 30º angle, a stance known as the Vertical de Barré position (Fig. 2). Once positioned, the subject was instructed to fix their gaze on a point at eye level and to remain silent whilst performing the Romberg test with eyes open (REO) for 30 s. Immediately afterwards, the subject repeated the same procedure, but this time with eyes closed, thus performing the Romberg test with eyes

Fig. 1 Dinascan/IBV® P600 Platform. *A real photograph taken during the study by the research team*

Fig. 2 Vertical Barré position for the Romberg TEST. *A real photograph taken during the study by the research team*

closed (REC) for another 30 s. The measured variables were: swept area in the Romberg test with eyes open (SAREO), swept area in the Romberg test with eyes closed (SAREC), medio-lateral displacement in the Romberg test with eyes open (MLDREO), medio-lateral displacement in the Romberg test with eyes closed (MLDREC), anteroposterior displacement in the Romberg test with eyes open (APDREO), anteroposterior displacement in the Romberg test with eyes closed (APDREC).

Quality of life assessment

Participants' quality of life was assessed using the Short-Form Health Survey (SF-36) [14] and the World Health Organisation Quality of Life Questionnaire (WHOQOL-BREF) [15]. Both questionnaires lack predetermined cutoff scores, meaning that higher scores indicate a better quality of life.

Variables and statistical analysis

A total of 18 variables were analysed: group (CrossFit or sedentary), sociodemographic (sex and age), anthropometric (weight, height, and BMI), ground reaction forces (average step speed, average support time, braking force, take-off force, propulsion force, and oscillation force), balance (swept area, medio-lateral displacement, and anteroposterior displacement) and quality of life (SF-36 and WHOQOL-BREF).

Amongst these, only "group" and "sex" were considered qualitative variables, whilst the rest were analysed as quantitative variables.

The statistical analysis was performed using JASP software for Windows, version 0.18.3. Normality was assessed using the Shapiro–Wilk test. Chi-square tests were used to analyse relationships between qualitative variables. *T* tests (Student's *T* test or Welch's test) were applied to compare quantitative variables between two groups, depending on Levene's test for homogeneity of variances. Paired *T* tests were used for comparisons of paired variables. In addition, Cramer's V and Cohen's d were calculated to evaluate the magnitude of the effect.

Finally, statistical power was estimated using the pwr package in R, version 4.5.1.

A p value < 0.05 was considered statistically significant for all analyses.

Results

Regarding sex distribution across the groups analysed, no statistically significant differences were found ($p = 0.243$), with a small effect size (Cramér's $V = 0.151$) and low statistical power (power = 0.216). On the other hand, concerning participants' age, statistically significant differences were observed between groups, with the CrossFit group being older on average ($p = 0.001$), and a large effect size (Cohen's $d = 1.182$), indicating a substantial practical difference in age between CrossFit and sedentary participants.

As for anthropometric variables such as weight, height and BMI, no significant differences were found between groups ($p \geq 0.101$). Nevertheless, the effect sizes suggest small to moderate differences in weight (Cohen's $d = 0.160$), height (Cohen's d = 0.430) and BMI (Cohen's $d = 0.279$) (Table 2).

In the analysis of ground reaction forces, no significant differences were found based on the foot analysed ($p \geq 0.351$), with a negligible effect size (Cohen's $d \geq 0.009$) and very low statistical power (power ≥ 0.050); therefore, the right foot was used for subsequent analyses. Descriptively, the CrossFit group showed higher values in average step speed, average support time and average braking force, whilst the sedentary group exhibited higher values in average propulsion force, average take-off force and average oscillation force. The only variable with a p value < 0.05 was average support time, which showed a statistically significant difference ($p = 0.040$) and a moderate effect size (Cohen's $d = 0.543$), indicating a slightly longer contact time in the CrossFit group (Table 3).

Regarding balance variables, participants in the CrossFit group demonstrated higher values in swept area, medio-lateral displacement and anteroposterior displacement under

Table 2 Relationship between sociodemographic and anthropometric variables by group *(Student's T test and Welch test)*

Variables	Group	Mean	Standard deviation	*p* value	Cohen's d	Power
Age (years)	Sedentary	26.00	1.72	**< 0.001***	1.182	0.994
	CrossFit	29.10	3.28			
Weight (Kg)	Sedentary	72.36	18.13	0.539	0.160	0.094
	CrossFit	69.62	16.12			
Height (cm)	Sedentary	164.43	7.40	0.101	0.430	0.374
	CrossFit	167.62	8.97			
BMI (Kg/m^2)	Sedentary	21.97	5.36	0.284	0.279	0.186
	CrossFit	20.62	4.20			

Kg kilogram, *cm* centimetres, *m*^{2} square metres, *BIM* body mass index,

*Statistically significant difference

Table 3 Relationship between ground reaction forces by group (*Student's T test and Welch test*)

Variables	Group	Mean	Standard deviation	*p* value	Cohen's d	Power
ASS (m/s)	Sedentary	1.12	0.09	0.773	0.075	0.059
	CrossFit	1.13	0.13			
AST (s)	Sedentary	0.70	0.05	**0.040***	0.543	0.543
	CrossFit	0.73	0.08			
ABF (N)	Sedentary	97.29	23.70	0.761	0.079	0.060
	CrossFit	98.17	38.59			
APF (N)	Sedentary	128.27	32.20	0.604	0.135	0.081
	CrossFit	125.84	36.93			
ATF (N)	Sedentary	772.75	175.41	0.564	0.150	0.088
	CrossFit	763.92	169.03			
AOF (N)	Sedentary	574.96	168.40	0.700	0.100	0.067
	CrossFit	555.08	138.77			

ASS average step speed, *m/s* metres per second, *s* seconds, *AST* average support time, *ABF* average braking force, *N* Newton, *APF* average propulsion force, *ATF* average take-off force, *AOF* average oscillation force

*Statistically significant difference.

the Romberg test with eyes open. Conversely, the sedentary group presented higher values with eyes closed. The anteroposterior displacement in the Romberg test with eyes open (APDREO) variable showed a statistically significant difference ($p = 0.019$) with a moderate effect size (Cohen's $d = 0.632$), being greater in the CrossFit group. The remaining variables did not show significant differences, although some effect sizes were moderate (Cohen's d ≈ 0.4) (Table 4).

Finally, concerning quality of life, statistically significant differences in favour of the CrossFit group were observed ($p \leq 0.009$) in both the WHOQOL-BREF and SF-36 questionnaires, with a large effect size (Cohen's $d \geq 0.701$) and high statistical power (power ≥ 0.761). This suggests that physically active participants perceive a better overall quality of life.

Although several comparisons did not reach statistical significance, it should be noted that, with the exception of age (power = 0.99), post hoc power analyses indicated low to moderate values (range: 0.059 to 0.700). This suggests a considerable risk of Type II error, meaning that real differences between the groups may not have been detected due to the small sample size. For this reason, effect sizes (Cohen's d) were also calculated, as they allow for the assessment of whether differences, although not statistically significant, might have practical or clinical relevance.

Discussion

The present study explored the effect of CrossFit training on gait biomechanics, postural balance and quality of life. Although participants in the CrossFit group showed descriptively higher values in average step speed, average support time and average braking force, these differences were not statistically significant. This lack of significance may be explained by factors such as the small sample size, the heterogeneity of the participants (in terms of experience,

Table 4 Relationship between balance variables by group (*Student's T test and Welch test*)

Variables	Group	Mean	Standard deviation	*p* value	Cohen's d	Power
SAREO (mm^2)	Sedentary	42.00	22.55	0.153	0.374	0.297
	CrossFit	52.18	31.17			
SAREC (mm^2)	Sedentary	53.71	37.09	0.139	0.387	0.314
	CrossFit	41.87	22.20			
MLDREO (mm)	Sedentary	13.81	4.17	0.136	0.392	0.321
	CrossFit	16.07	7.00			
MLDREC (mm)	Sedentary	14.36	5.08	0.948	0.017	0.050
	CrossFit	14.28	5.10			
APDREO (mm)	Sedentary	17.97	5.50	**0.019***	0.632	0.673
	CrossFit	24.47	13.46			
APDREC (mm)	Sedentary	22.65	7.60	0.362	0.237	0.147
	CrossFit	20.96	6.64			

SAREO swept area in the Romberg test with eyes open, *mm^2* square millimetres, *SAREC* swept area in the Romberg test with eyes closed, *MLDREO* medio-lateral displacement in the Romberg test with eyes open, *MLDREC* medio-lateral displacement in the Romberg test with eyes closed, *APDREO* anteroposterior displacement in the Romberg test with eyes open, *APDREC* anteroposterior displacement in the Romberg test with eyes closed

*Statistically significant difference.

frequency, and training intensity) or the insufficient duration of stimulus to induce measurable adaptations in these variables.

These findings are consistent with recent reviews that have critically examined the impact of CrossFit on health and physical performance. Claudino et al. [16] pointed out that most studies on CrossFit suffer from methodological limitations, including cross-sectional designs and small samples, which restrict the external validity of the findings and the ability to draw firm conclusions regarding its physiological or biomechanical benefits. More recently, Martinho et al. [17] conducted a systematic scoping review including 68 original studies on CrossFit in adults and concluded that sport-specific assessments (such as strength and power in functional exercises) are more effective at predicting performance than general variables such as VO_2max. Moreover, they highlighted that most available research lacks a longitudinal design, and that a recovery period of at least 48 to 72 h is necessary following intense sessions to correctly interpret training responses. This review also identifies gaps in the literature concerning the effects of CrossFit on aspects such as functional stability or gait biomechanics, thereby reinforcing the need for targeted studies in these areas [17].

Concerning balance, a significantly greater anteroposterior sway was observed in the CrossFit group during the eyes-open Romberg test. This may reflect muscular adaptations due to the dominance of sagittal-plane movements (e.g. squats, deadlifts, Olympic lifts) in CrossFit training. However, this finding does not necessarily imply improved postural control and should be interpreted cautiously. In the eyes-closed Romberg condition, although the CrossFit group exhibited descriptively better scores, no statistically significant differences were found. These results do not support robust proprioceptive advantages, though they could be hypothesised and investigated in future research using dynamic functional tests and tighter control of training variables.

Regarding quality of life, participants in the CrossFit group reported significantly higher scores compared to the sedentary group. These findings align with previous research documenting the positive impact of CrossFit on psychological well-being, motivation and social engagement. For example, Box et al. [18] observed that CrossFit practitioners exhibit elevated levels of intrinsic motivation, enjoyment and social affiliation compared to participants of other modalities. Similarly, Bycura et al. [19] surveyed over 700 CrossFit practitioners and found that intrinsic motives, such as enjoyment, challenge and a sense of community, correlated strongly with both frequency and duration of participation, supporting adherence and perceived well-being. In addition, Ficarra et al. [20] showed that combining CrossFit training with a Mediterranean diet led to notable improvements in body composition and physical performance amongst trained adults, suggesting that comprehensive lifestyle interventions may enhance the benefits of functional training. However, the present study did not control for diet or other contextual factors, which limits the direct application of these findings.

Amongst the main limitations of the present study is its cross-sectional design, which precludes establishing causal relationships between CrossFit practice and the analysed variables. In addition, the sample size was small and unbalanced in terms of sex, age and training experience, limiting the generalisability of the results and reducing the statistical power of the analyses. Sample heterogeneity, including differences in training frequency, duration, intensity and general lifestyle, may have contributed to variability in the outcomes. Furthermore, the tests used to assess gait and balance were mostly static or linear, which may not adequately capture the complexity of dynamic functional performance that characterises CrossFit. Potentially influential external variables such as diet, sleep or stress levels were not controlled, introducing possible confounding factors. Finally, participant self-selection may have introduced biases related to motivation or subjective perceptions of quality of life, further limiting the external validity of the study.

Conclusion

Although no statistically significant differences were found in most biomechanical variables, individuals who practise CrossFit reported a better perceived quality of life and showed trends in postural balance that warrant further attention. Nevertheless, the exploratory nature of this study and its limited statistical power require that these results be interpreted with caution. No solid evidence was found of improvements in gait kinetics. Longitudinal studies with larger and more balanced samples, dynamic functional testing, and greater control of contextual variables will be necessary to confirm or refute these potential benefits.

Acknowledgements The authors would like to thank all the voluntary participants, without whom this study would not have been possible.

Author contributions O.L.R and R.M.A. wrote the main manuscript text and J.V.R and C.M.F. prepared all figures. All authors reviewed the manuscript.

Funding Open Access funding provided thanks to the CRUE-CSIC agreement with Springer Nature. This work was carried out within the framework of the pre-doctoral contract funded by the Regional Government of Extremadura through the call approved by DECREE 43/2023, of May 3, which establishes the regulatory bases for granting aid for the recruitment of pre-doctoral research personnel in training within the Extremadura System of Science, Technology, and Innovation, and approves the first call (2023040085). This contract is co-financed by the European Union through the European Social Fund Plus (ESF+), within the operational programme for Extremadura, contributing to the strengthening of Page containing contact details human capital in R&D&I and to the strategic objectives of the European Union for the promotion of scientific excellence and innovation. This study was funded by the Regional Government of Extremadura and the European Regional Development Fund (ERDF) through a grant to the research group [code CTS020, reference GR24135].

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Informed Consent The study was approved by the Bioethics and Biosafety Committee of the University of Extremadura (Spain), under registration number 199/2023, which supports its ethical validity Also in line 107 It Talk about informed consent signed by all participants. All included in materials and methods section.

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